Worksheet 2.1 — Function Basics: Domain, Range and Notation

Mathematics: Analysis & Approaches SL (IB) · Unit 2: Functions

The Big Ideas

A function assigns exactly one output to every input; its domain and range describe what inputs and outputs are possible.

- Domain restrictions: avoid division by zero and square roots of negatives.
- Vertical line test: a graph is a function iff no vertical line crosses it more than once.
- Range: think about the smallest and largest outputs the function can actually reach.

Worked Examples

Example 1 — Evaluate a function

Given $f(x) = 3x^2 - 4x + 1$, find $f(-2)$.
Solution. $3(4) - 4(-2) + 1 = 12 + 8 + 1$.
Conclusion: $f(-2) = 21$.

Example 2 — Domain with a root

Find the domain of $f(x) = \sqrt{2x - 8}$.
Solution. $2x - 8 \geq 0 \Rightarrow x \geq 4$.
Conclusion: domain is $x \geq 4$.

Example 3 — Domain with a fraction

Find the domain of $f(x) = \dfrac{5}{x^2 - 9}$.
Solution. $x^2 - 9 \neq 0 \Rightarrow x \neq \pm 3$.
Conclusion: all real x except 3 and -3.

Example 4 — Range from a transformed root

Find the range of $f(x) = -\sqrt{x - 2} + 5$.
Solution. Domain is $x \geq 2$; at $x = 2$, $f = 5$ (largest value, since $-\sqrt{\ }$ falls as x grows).
Conclusion: range is $f(x) \leq 5$.

Example 5 — Combined domain restrictions

Find the domain of $f(x) = \dfrac{\sqrt{x + 2}}{x - 3}$.
Solution. Need $x + 2 \geq 0 \Rightarrow x \geq -2$, and also $x - 3 \neq 0 \Rightarrow x \neq 3$.
Conclusion: domain is $x \geq -2$, $x \neq 3$.

Example 6 — Range on a restricted domain

Find the range of $f(x) = x^2 - 4x + 3$ for $0 \le x \le 5$.

Solution. **Step 1:** Vertex at $x = 2$ (inside the domain): $f(2) = 4 - 8 + 3 = -1$ (minimum).

Step 2: Check endpoints: $f(0) = 3$, $f(5) = 25 - 20 + 3 = 8$.

Conclusion: range is $-1 \le f(x) \le 8$.

Example 7 — Application — domain from a real constraint

A rectangle's area is $A(x) = x(20 - x)$, where x is one side and $20 - x$ is the other (from a fixed perimeter). State the domain that makes physical sense.

Solution. Both sides must be positive: $x > 0$ and $20 - x > 0$.

Conclusion: domain is $0 < x < 20$.

Example 8 — Testing a set of ordered pairs

Determine whether $\{(1, 2), (2, 3), (1, 5)\}$ represents a function.

Solution. The input $x = 1$ appears twice, with two different outputs (2 and 5).

Conclusion: not a function — one input must give exactly one output.

Example 9 — Range of a semicircle function

Find the range of $f(x) = -\sqrt{16 - x^2}$ (domain $-4 \le x \le 4$).

Solution. This is the lower half of a circle of radius 4; f ranges from -4 (at $x = 0$) up to 0 (at $x = \pm 4$).

Conclusion: range is $-4 \le f(x) \le 0$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Given $f(x) = 2x^2 + 5x - 1$, find $f(3)$.

Question 2

Find the domain of $f(x) = \sqrt{5 - x}$.

Question 3

Find the domain of $f(x) = \dfrac{4}{2x + 6}$.

Question 4

Find the domain of $f(x) = \dfrac{1}{\sqrt{x - 1}}$ (careful: the root cannot be zero either).

Question 5

Find the range of $f(x) = \sqrt{x + 3} - 1$.

Question 6

Determine whether $x^2 + y^2 = 25$ passes the vertical line test, and explain your reasoning.

Question 7

Find the domain of $f(x) = \dfrac{x+2}{x^2-4}$ (factor the denominator first).

Question 8

Find the domain of $f(x) = \dfrac{\sqrt{x-1}}{x-5}$.

Question 9

Find the range of $f(x) = x^2 + 2x - 3$ for $-3 \leq x \leq 2$.

Question 10

Determine whether $\{(2,4),(3,4),(2,7)\}$ represents a function, and explain.

Question 11 — Challenge

Find the range of $f(x) = -2\sqrt{x} + 7$.

Question 12 — Challenge

Find the range of $f(x) = \sqrt{25 - x^2}$.

Question 13 — Challenge

A tank's volume model is $V(t) = 100 - 4t$ litres for a draining tank starting at 100 L; state the domain that makes physical sense.

Question Recap & Answer Key

Question Recap

1. Given $f(x) = 2x^2 + 5x - 1$, find $f(3)$.
2. Find the domain of $f(x) = \sqrt{5 - x}$.
3. Find the domain of $f(x) = \dfrac{4}{2x + 6}$.
4. Find the domain of $f(x) = \dfrac{1}{\sqrt{x - 1}}$ (careful: the root cannot be zero either).
5. Find the range of $f(x) = \sqrt{x + 3} - 1$.
6. Determine whether $x^2 + y^2 = 25$ passes the vertical line test, and explain your reasoning.
7. Find the domain of $f(x) = \dfrac{x + 2}{x^2 - 4}$ (factor the denominator first).
8. Find the domain of $f(x) = \dfrac{\sqrt{x - 1}}{x - 5}$.
9. Find the range of $f(x) = x^2 + 2x - 3$ for $-3 \leq x \leq 2$.
10. Determine whether $\{(2, 4), (3, 4), (2, 7)\}$ represents a function, and explain.
11. Find the range of $f(x) = -2\sqrt{x} + 7$.
12. Find the range of $f(x) = \sqrt{25 - x^2}$.
13. A tank's volume model is $V(t) = 100 - 4t$ litres for a draining tank starting at 100 L; state the domain that makes physical sense.

✓ Answer Key

1. $18 + 15 - 1 = 32$
2. $x \leq 5$
3. $x \neq -3$
4. $x > 1$
5. $f(x) \geq -1$
6. Fails: a vertical line at $x = 0$ meets the circle at both $y = 5$ and $y = -5$
7. $x^2 - 4 = (x - 2)(x + 2)$; domain is all real x except 2 and -2
8. need $x \geq 1$ and $x \neq 5$: domain is $x \geq 1, x \neq 5$
9. vertex at $x = -1$ (inside domain): $f(-1) = -4$ (min); endpoints $f(-3) = 0$, $f(2) = 5$; range $-4 \leq f(x) \leq 5$
10. Not a function — $x = 2$ gives two different outputs, 4 and 7
11. domain $x \geq 0$; at $x = 0$, $f = 7$ (max); as x grows, $f \to -\infty$: range $f(x) \leq 7$
12. upper half of a circle radius 5; range $0 \leq f(x) \leq 5$
13. tank can't hold negative volume or run past empty: $0 \leq t \leq 25$